

Seizure

- Seizure V.S. Epilepsy

Epilepsy	Seizure
A chronic neurological disorder	A sudden, abnormal surge of electrical activity in the brain
Characterised by repetitive seizures	Many factors, such as head injuries, brain infections, and hereditary predisposition can set it
Needs constant medical attention	Medications, operations, and other treatments may be used to alleviate symptoms
Symptoms may include altered behaviour, motor skills, or state of consciousness.	Can differ in both its occurrence and its degree of impact

  - Epilepsy
    - >2 unprovoked seizures/reflex seizures >24 hr interval
    - 1 unprovoked/reflex seizure and a probability of further seizures similar to the general recurrence risk (> 60%) after two unprovoked seizures, occurring over the next 10 years
    - diagnosis of an epilepsy syndrome
- Seizure causes: CDEF, GHI

  CNS structure lesion: 先天性腦部結構異常, 腦瘤, 腦部出血

  Drug/Intoxication: 藥物中毒, 忘記服用癲癇藥物

  Electrocyte: hyper/hyponatremia

  Encephalopathy: Hepatic encephalopathy, Uremic encephalopathy, Inborn error of metabolism

  Febrile seizure

Glucose: hypoglycemia

Hypoxia

Infection/Inflammation: meningitis, encephalitis, SLE related CNS lupus

- Hx taking: 4P

  Predisposing factor: 打電動, fever

  Prodromal sign

  Past hx

  Pattern: AIDP+F

- 如何觀察抽搐的發作: AIDP+F

  Aura: 有無先兆(發呆, 大哭, 大叫)

  Ictal: 留意觀察發作情形(全身/局部), 型態:眼→嘴→四肢→大小便失禁

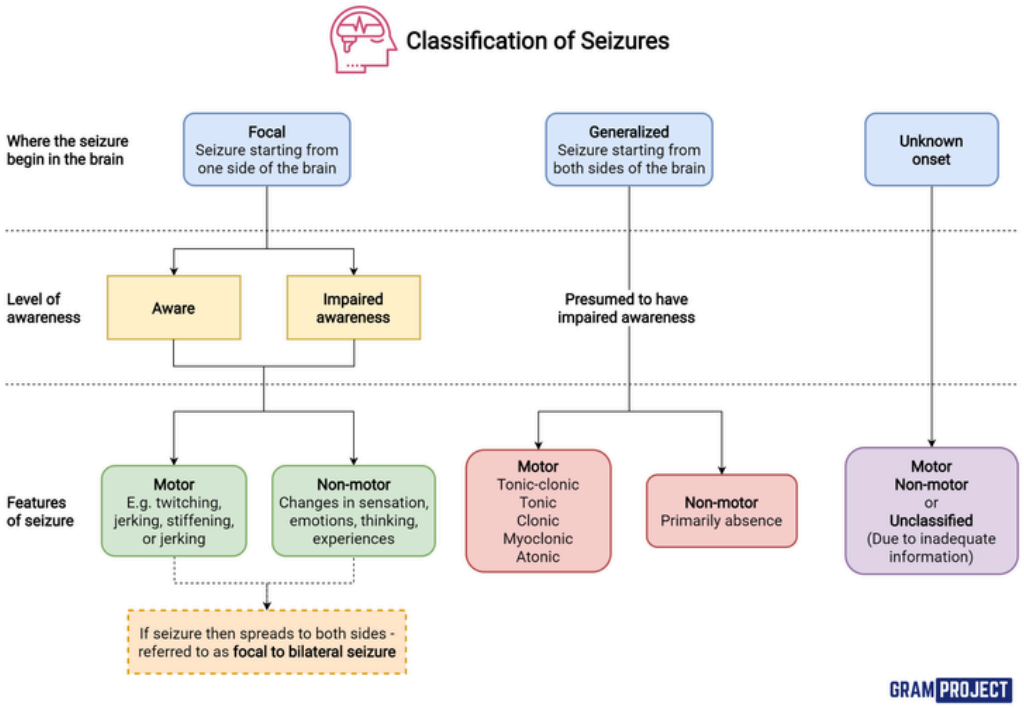
  Duration: 發作時長

  Post-ictal: 發作後睡覺or醒著

  Frequency: 發作次數

- Classification:

  1. Generalized tonic-clonic type 大多數
  2. Tonic type
  3. Clonic type
  4. Atonic type
  5. Focal motor type 不典型, 轉變成癲癇可能性較高



Febrile Seizures

<D/D> Fever/Chillness:
1. Febrile seizure 會意識昏迷
2. 握緊孩童的手發燒會停止, 熱性痙攣仍然抽個不停

define	2-5% incidence, in child, usually 6M-6Y (peak 1.5Y) 因為發燒出現的抽搐(需排除中樞神經感染及其他已知疾病)
causes/ rf	1. 腦部發育未成熟 2. Fever(溫度高低&progress) 3. 家族性遺傳
s/s	- 開始發燒後24H內 - 大發作: 全身性身體的抽動, lips cyanosis, conscious change, 眼球上吊 - 第一次發作後約有三分之一會第二次發作 Classification: 1. **Simple febrile seizure**: 第一天發燒內一次發作<15min且呈現GTC type 2. **Complex febrile seizure**: >15min, >2times a day, 局部發作,發作後仍可發

	現局部症狀 3. **Febrile status epileptics**: >30min
	Genetic Epilepsy w/ febrile seizure plus: - 一開始以熱痙攣表現 - 六歲後還出現無發燒的抽搐 (全身性/局部性) - 與自體顯性遺傳相關 SCN1B, SCN1A, GABRG2, STX18 - 每個家族成員在癲癇表現和嚴重度上可能有很大的差異 - 大多數病患到青春期前通常會慢慢緩解
dx	重點在於發燒原因的檢查 1. Lumbar puncture: rule out meningitis 2. Electroencephalogram (EEG): for pt that is complex febrile seizure 3. Brain MRI: for pt that is complex febrile seizure or had past hx
mx	1. 維持呼吸道暢通 2. 取下身體約束物 3. 側躺以防嗆到 4. 清除現場危險物品
recurrence	1. Family Hx 2. 1st 發作年齡<18M 3. 發作時體溫較高, 復發機率較小 4. 發燒時間越短, 復發機率越高
tx	成長到5,6Y腦部發育成熟會自然消失 符合complex type febrile seizure 任一條件可給藥(有時會等2nd 同樣情形才給藥)

TABLE 1.

Classification of Febrile Seizures

	Simple Febrile Seizure (all criteria must be met)	Complex Febrile Seizure (one or more of the following)
Duration	Short (< 15 minutes) Self - limiting	Prolonged febrile seizure (> 15 minutes) Febrile status epilepticus (> 30 minutes)
Phenotype	Generalized tonic-clonic	Focal onset or features Clonic and/or tonic Loss of muscle tone Focal progressing to general Head/eye deviation to one side
Recurrence Frequency	No recurrence in 24 hours	Recurrence within the same febrile illness
Prior Neurologic Diagnosis	None	Present
Post-ictal Pathology	None	Present (unilateral paralysis, somnolence)

Partial Seizure

Definition: 大腦皮質某一區域有不正常電氣生理活動(不一定意識喪失)

Simple Partial Seizures (SPS)

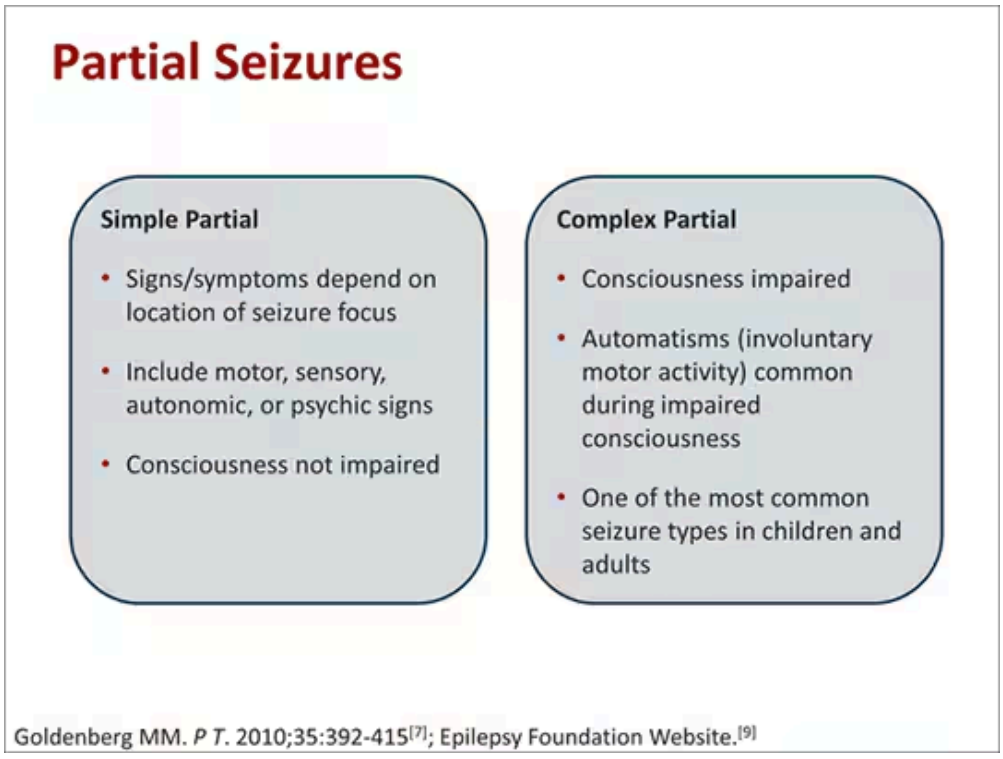
1. Motor activity: 臉、頸、四肢 tonic or clonic
2. Aura: 頭痛、胸部不舒服

Complex Partial Seizures (CPS)

1. Emotion and memory function (temporal lobe)
2. Aura
3. Altered consciousness

3. 保留意識和言語
4. 結束後不會出現其它症狀

4. 50～75% Automatism(年紀越大越易出現)
5. Secondary tonic-clonic convulsion: 發作過程中若有再發生不正常放電
6. Postictal period: tired, sleepy
7. EEG: 前顳葉有sharp waves, focal spikes, multifocal spikes (20% EEG normal)
8. CT/MRI: for CPS病灶(尤其顳葉)



Benign Partial Epilepsy with Centrotemporal Spikes (BPEC)

1. 2-14Y(特別9-10), very good prognosis
2. Rf: No past hx but has family hx, 75%在睡夢中
3. 僅侷限在臉部, motor signs & somatosensory symptoms
4. EEG: centrotemporal/rolandic can see repetitive spike focus

5. Oxcarbazepine (partial seizures首選藥物): 持續給藥>2Y or 到14-16Y

Generalized Seizure

Definition: 兩側大腦皮質同時發生不正常電氣生理活動, mostly lose consciousness

Absence Seizures

1. 小發作(petit mal)
2. 發作時動作靜止, 面無表情, 發作期間無法藉由呼叫或碰觸喚醒
3. <5Y 不常發生, 女性
4. No aura, <30sec
5. Hyperventilation 3-4min or 閃光 may trigger
6. 一天內可以發生很多次，甚至可達數十次
7. 90% 病人有 automatic behavior。
8. EEG: 3/sec spike, generalized wave discharge

Generalized Tonic-Clonic Seizures (GTC)

1. 大發作(grand mal)
2. 可能會有aura: partial seizure before GTC
3. Tonic phase: 全身性肌肉群收縮, apnea and cyanosis
   Clonic phase: 當肌肉收縮被反覆性陣發性肌肉鬆弛所阻斷
4. Lose sphincter control(bladder)

Myoclonic Epilepsy of Childhood

1. 短暫重複對稱性肌肉收縮, 同時發生 body tone loss
2. Subtype:
   a. Benign myoclonus of infancy: neck, trunk, extremities
      - normal EEG, good prognosis, 2Y前消失, no

Infantile Spasm

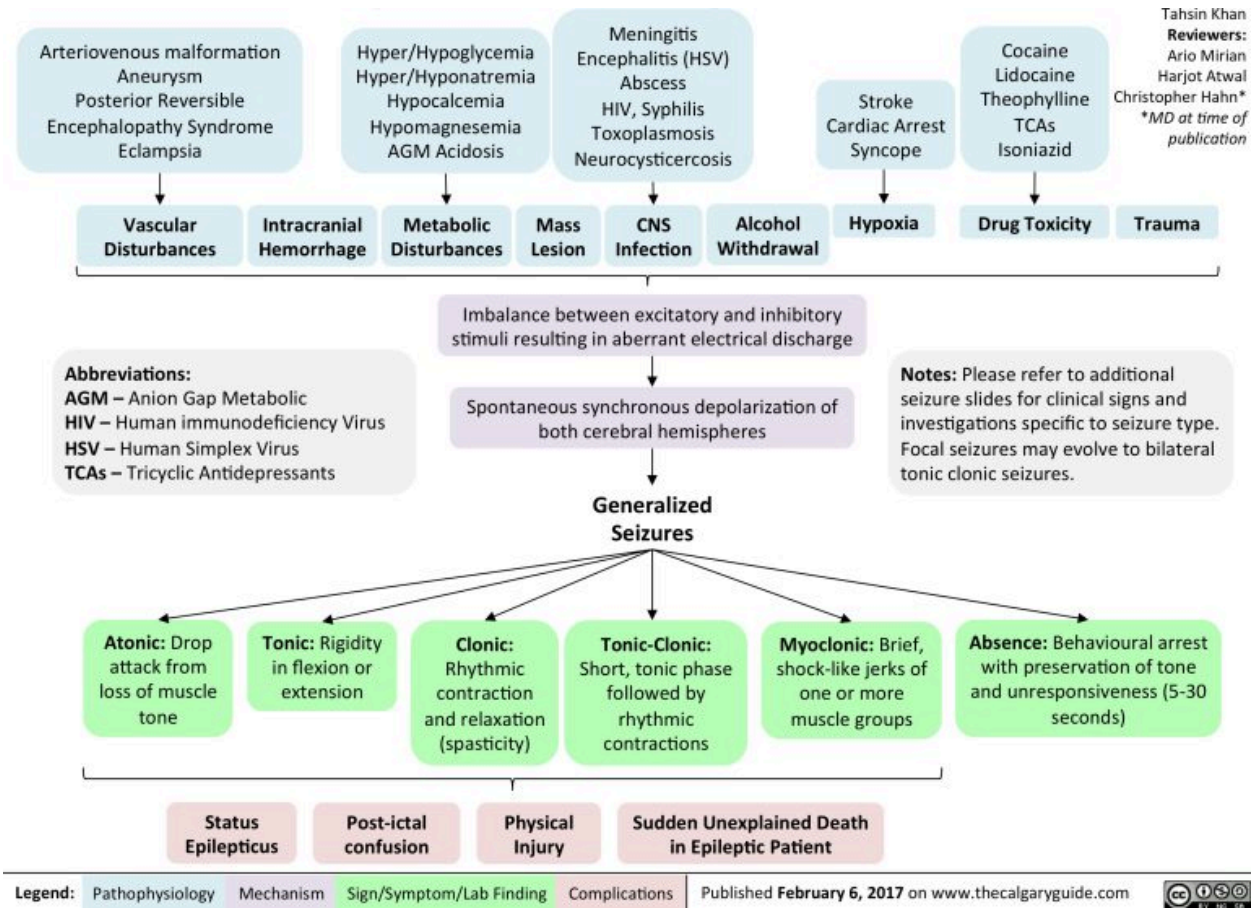
1. 4-8M, 頸部、軀幹與四肢有對稱性肌肉收縮, 剛睡覺/剛睡醒
2. Groups:
   a. Cryptogenic group: 10-20%
      - 發作之前normal developmental milestone, normal brain CT/MRI, good prognosis

need drug tx

b. Complex myoclonic epilepsies: bad prognosis, 1Y partial/generalized tonic-clonic seizures
    - If generalized, often associated with URI→status epilepticus, hx of hypoxic-ischemic encephalopathy (birth)

b. Symptomatic group:
    - 80-90% Intellectual disability, 與出生前/過程/後發生中樞神經疾病有關 (infection, hypoxic-ischemic encephalopathy with PVL, neurocutaneous syndromes such as tuberous sclerosis)

3. EEG: hypsarrhythmia
4. Tx: VitB6, antiepileptic drugs, ACTH, steroids, 生酮飲食有幫助



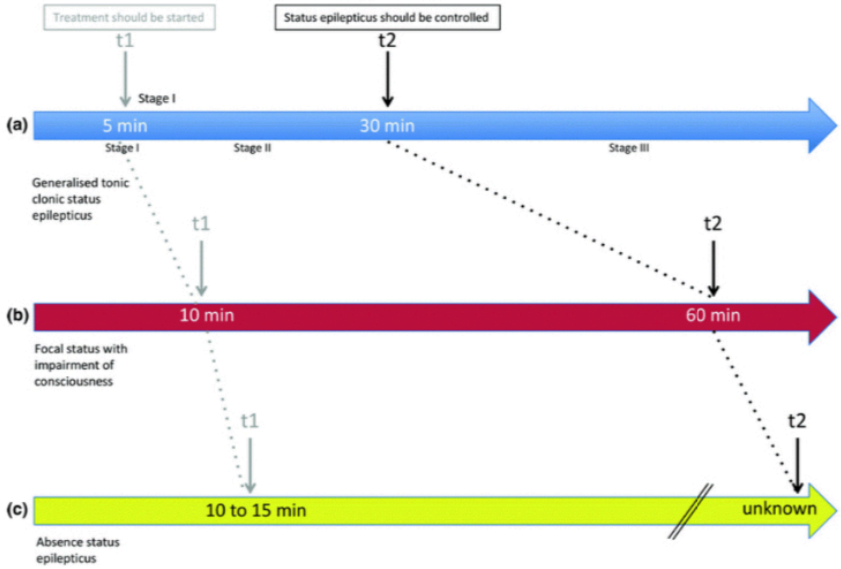
Dravet Syndrome

define	severe myoclonic epilepsy of infancy, developmental epileptic encephalopathy
causes/ rf	SCN1A gene mutation(80%), voltage-gated sodium channel (neuronal cell bodies & dendrites) Rf: Temperature change, fever, vaccination, infection
s/s	<1Y(4-8M), 發作前的神經學發育正常 May combined with Autism spectrum (不一定有speech delay) 第一次發作: prolonged partial seizure抽搐+發燒, Status epilepticus 1. Combination of various forms of epilpesy 2. Neurodevelopmental delay 3. Neurologic disability begin after the onset of seizure
dx	1. Hx/Rf, NE 2. Genetic test (不一定有SCN1A突變) 3. MRI (often normal) / EEG
tx	頑固型. 需要數種抗癲癇藥物同時治療, 儘早考慮使用生酮飲食 1. 可用: Valproate, Clobazam, Levetiracetam, Topiramate, Clonazepam, Stiripentol 2. 不可使用: Lamotrigine, Phenobarbital, Oxcarbazepine, Carbamazepine, Phenytoin (阻礙鈉離子通道)

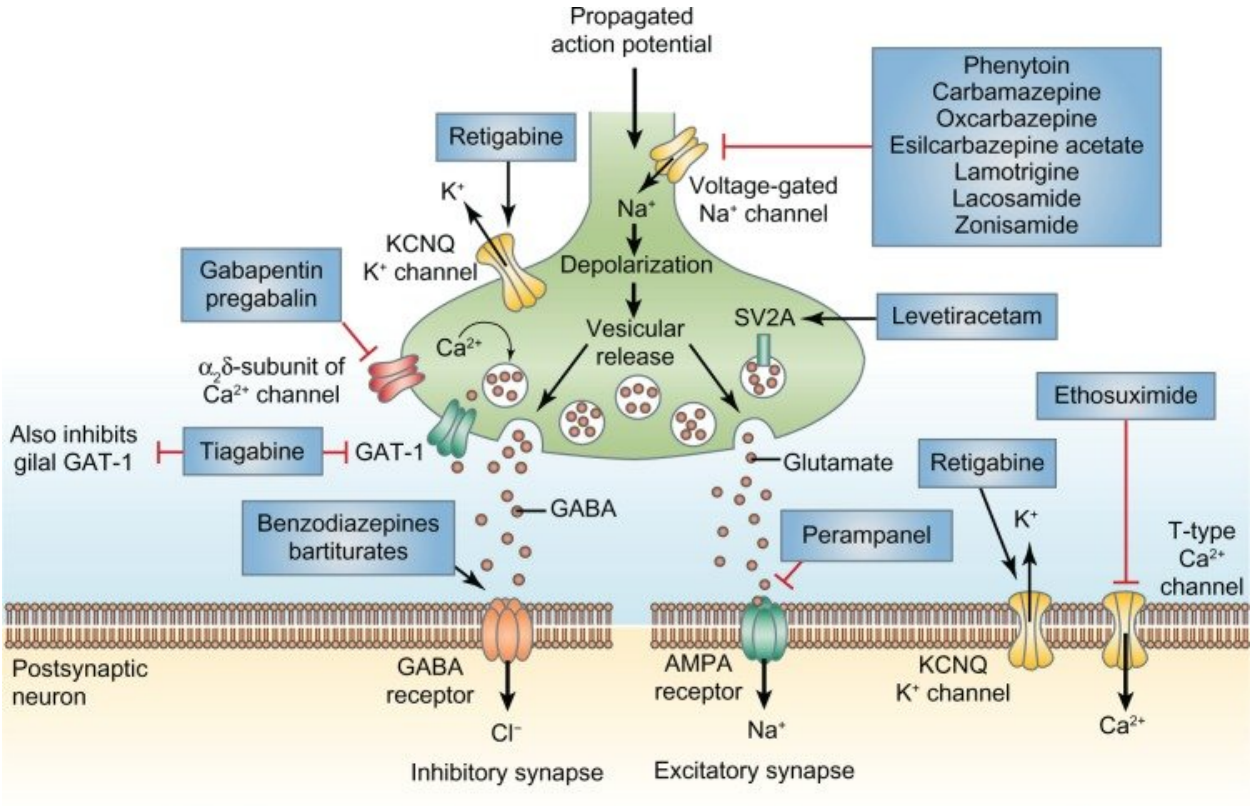
Status epilepsy

T1: 進展成status

T2: 造成不可逆腦傷

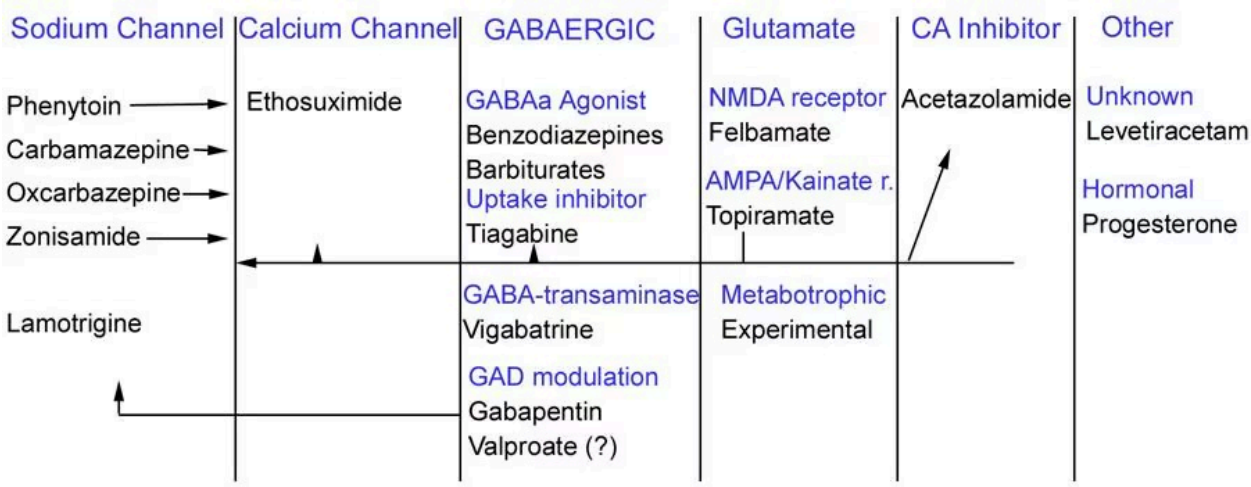


Anti-seizure medication



Not illustrated:

- Vigabatrin → ↓GABA degradation and drugs with multiple mechanisms:
- Valproate → ↑GABA turnover, ↓ Na^+ channels, ↓NMDA receptors
- Topiramate → ↓Na^+ channels, ↓AMPA/kainate receptors, ↑$GABA_A$ receptors
- Felbamate → ↓ Na^+ channels, ↑$GABA_A$ receptors, ↓NMDA receptors



HEPATIC ENZYME INDUCERS

PHENYTOIN
CARBAMAZEPINE (also autoinduction)
BARBITURATES
OXCARBAZEPINE
TOPIRAMATE (weak)

MAINLY RENALLY EXCRETED

GABAPENTIN
LEVETIRACETAM
TOPIRAMATE (lesser extend)

CONCOMITANT MIGRAINE

VALPROATE
GABAPENTIN
TOPIRAMATE

WEIGHT LOSS

TOPIRAMATE
ZONISAMIDE

PARENTERAL AVAILABLE

PHENYTOIN/FOSPHENYTOIN
VALPROATE
BARBITURATES
BENZODIAZEPINES

ONCE DAILY DOSE

PHENYTOIN
ZONISAMIDE
VALPROATE
PHENOBARBITAL

HIGH PROTEIN BINDING

PHENYTOIN	(70-90%)
VALPROATE	(85-95%)
TIAGABINE	(96%)
CARBAMAZEPINE	(75%)
CLOBAMAZEPINE/CLONAZEPAM	(83%-86%)
PHENOBARBITAL	(45%-60%)

ACTIVE METABOLITIES

CARBAMAZEPINE	EPOXIDE
CLOBAZAM	N-DESMETHYLCLOBAZAM
OXCARBAZEPINE	10-MONOHYDROXY (MHD)
PRIMIDONE	PHENOBARNITAL

AVOID IN YOUNG WOMEN

VALPROATE	higher teratogenic risk
PHENYTOIN	cosmetic effects, hirutism

MANAGEMENT OF CLUSTER SEIZURES

LORAZEPAM PERORALLY	0.03-0.05mg/Kg
RECTAL DIAZEPAM GEL	0.2-0.5mg/Kg

• 病房常用抗癲癇藥物簡表（2015）

藥物名稱	中文	主要劑型	病房常用劑量	主要代謝路徑
Phenobarbital (luminol)	魯米那	100mg/ml/amp 30mg/tab(PO)	Loading: 10-15mg/kg Maintain: 3-5mg/kg/D	肝臟 腎臟
Levetiracetam (keppra)	優癇	100mg/ml/amp 500mg/tab(PO) 100mg/ml(Syrup)	Loading:10-20mg/kg Maintain:10-50mg/kg/D	腎臟
Valproic acid (depakin)	帝拔癲	400mg/vial 200mg/tab(PO) 500mg/tab(PO) 200mg/ml(Syrup)	Loading:10-20mg/kg Maintain:20-40mg/kg/D	肝臟
Phenytoin (dilantin)	癲能停	250mg/5ml/amp 100mg/cap(PO)	Loading:10-15mg/kg Maintain:5mg/kg/D	肝臟
Topiramate (topamax)	妥泰	25mg/tab(PO) 100mg/tab(PO)	1-9mg/kg/D	腎臟
Oxcarbazepine (trileptal)	除癲達	300mg/tab(PO) 60mg/ml(Syrup)	10-60mg/kg/D	肝臟
Zonisamide	佐能安	100mg/tab(PO)	2-4mg/kg/D	肝臟
Clonazepam (rivotril)	利福全	0.5mg/tab(PO) 2mg/tab(PO)	0.1-0.2mg/kg/D	肝臟分解； 腎臟排泄
Clobazam	服利寧	10mg/tab(PO)	起始劑量5mg/D	肝臟分解； 腎臟排泄
Lamotrigine (lamictal)	樂命達	100mg/tab(PO)(高) 50mg/tab(PO)(中) 5mg/tab(chewable)(低)	視其他抗癲癇藥之交互作用而定	肝臟
Vigabatrin (sabril)	赦癲易	500mg/tab(PO)	30-100mg/kg/D	腎臟